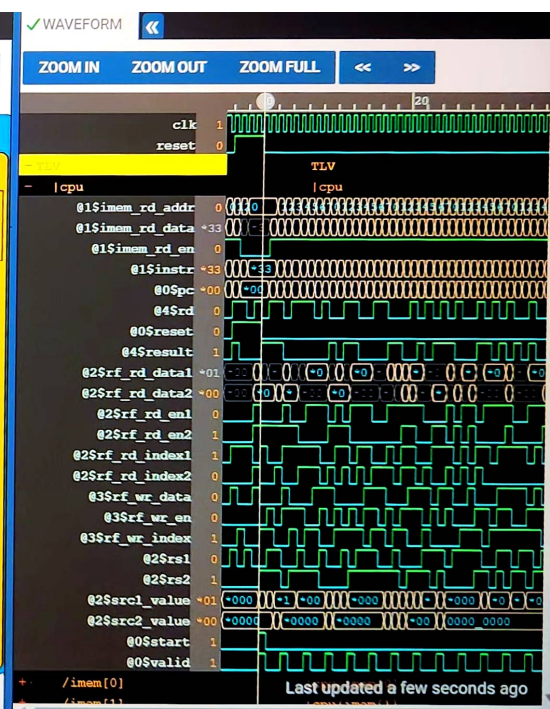
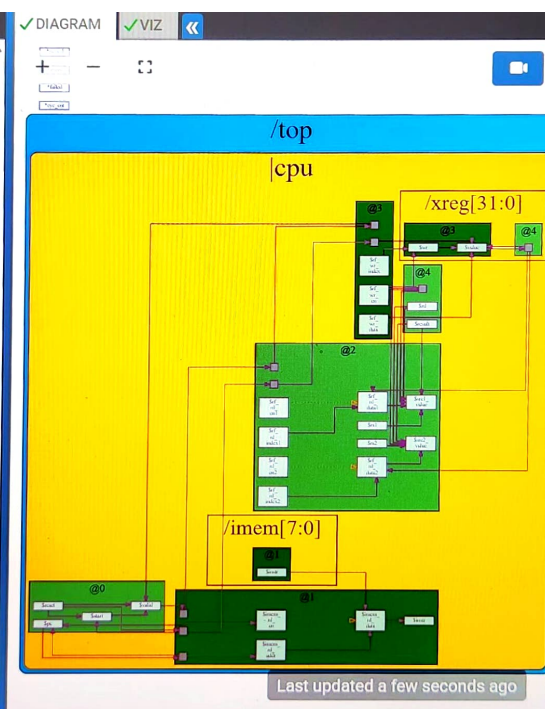


```

47 * $pc[31:0] = >>1$reset ? 32'd0 :
48 * >>1$pc + 32'd4;
49
50 @1
51 $mem_rd_en = !$reset;
52
53 $mem_rd_addr[M4_IMEM_INDEX_CNT-1:0] =
54 $pc[M4_IMEM_INDEX_CNT+1:2];
55
56 $instr[31:0] = $mem_rd_data;
57
58 @2
59 $src1_value[31:0] = (>>2$rf_wr_en && (
60 I
61 $src2_value[31:0] = (>>2$rf_wr_en && (
62
63
64
65 *passed = *cyc_cnt > 40;
66 *failed = 1'b0;
67
68 //cpu
69 m4+imem(@1)
70 m4+rf(@2, @3)
71 //m4+dmem(@4)
72
73 //m4+cpu_viz(@4)

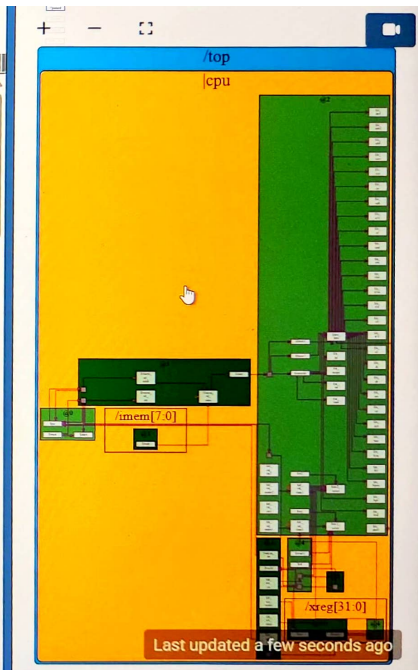
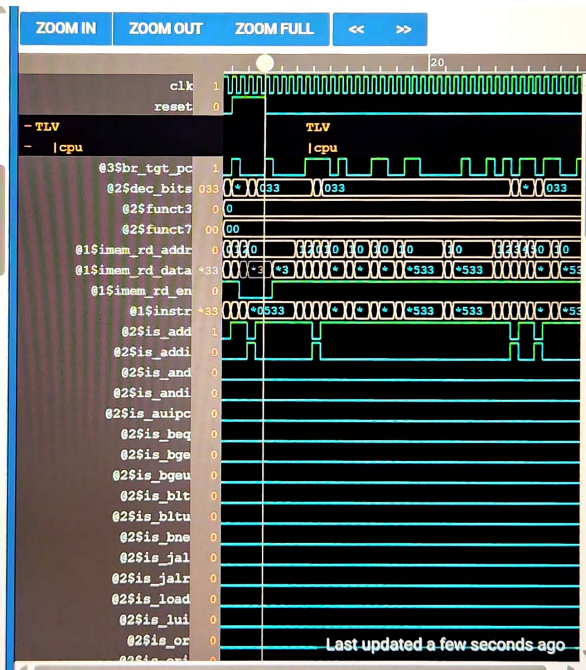
```



```

37
38
39 0
40 $reset = *reset;
41
42 $start = !$reset && >>1$reset;
43
44 $pc[31:0] =
45     >>1$reset    ? 32'd0 :
46     >>3$taken_br ? >>3$br_tgt_pc :
47                 >>1$pc + 32'd4;
48
49 1
50 $imem_rd_en = !$reset;
51
52 $imem_rd_addr[M4_IMEM_INDEX_CNT-1:0] =
53     $pc[M4_IMEM_INDEX_CNT+1:2];
54
55 $instr[31:0] = $imem_rd_data;
56
57 2
58
59 $src1_value = (>>2$rf_wr_en && (>>2$rd == $rs1)) ?
60
61 $src2_value = (>>2$rf_wr_en && (>>2$rd == $rs2)) ?
62
63

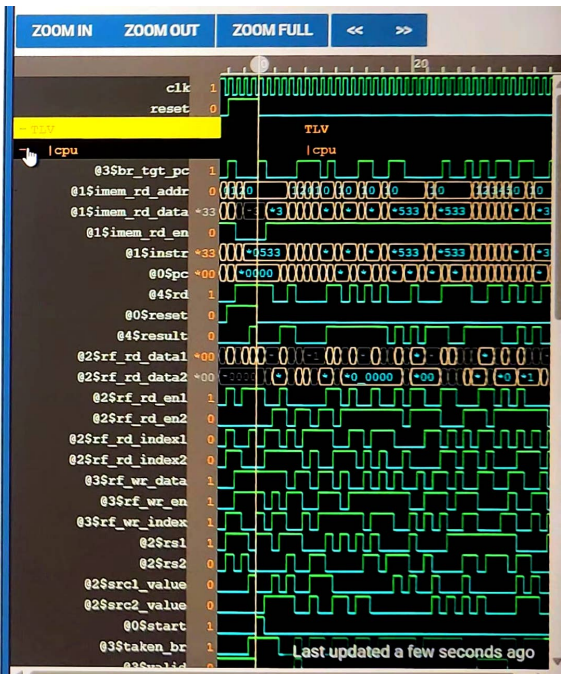
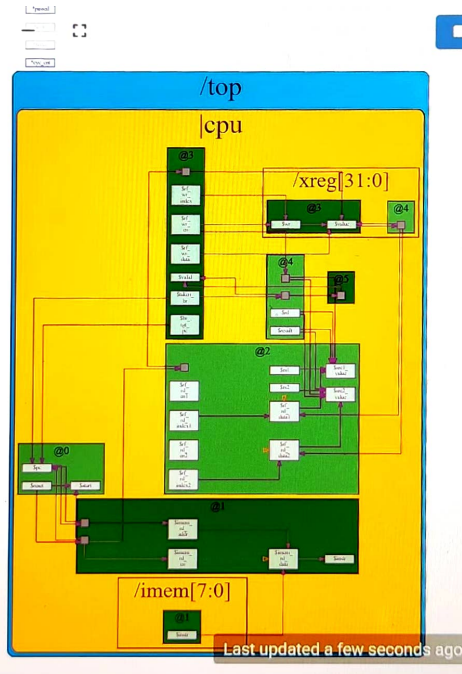
```



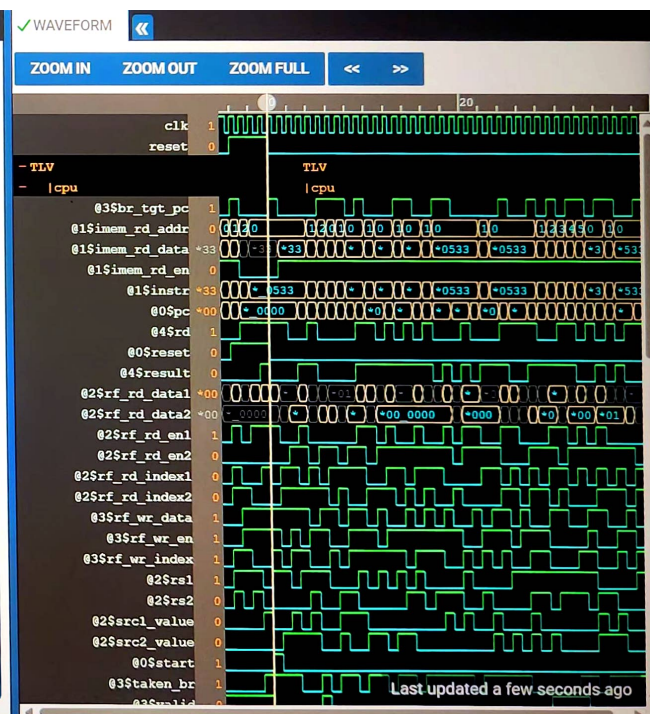
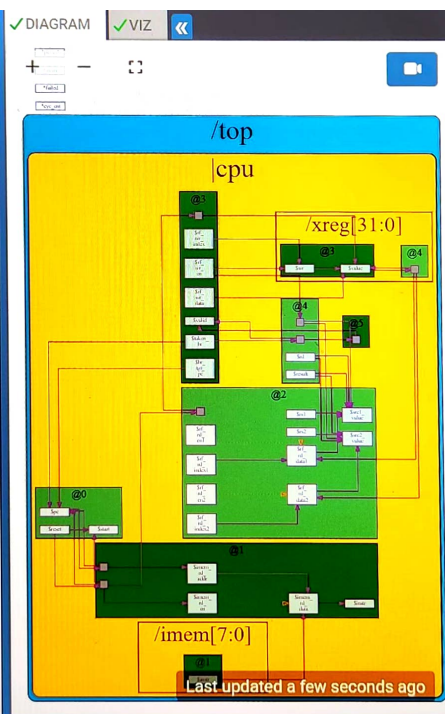
```

42 $start = !$reset && >>1$reset;
43
44 $pc[31:0] =
45     >>1$reset      ? 32'd0 :
46     >>3$taken_br   ? >>3$br_tgt_pc :
47     >>1$pc + 32'd4;
48
49 @1
50 $mem_rd_en = !$reset;
51
52 $mem_rd_addr[M4_IMEM_INDEX_CNT-1:0] =
53     $pc[M4_IMEM_INDEX_CNT+1:2];
54
55 $instr[31:0] = $mem_rd_data;
56
57 @2
58 $src1_value = (>>2$rf_wr_en && (>>2$rd
59 $src2_value = (>>2$rf_wr_en && (>>2$rd
60
61
62
63
64 @3
65 $valid = !(>>1$taken_br || >>2$taken_b
66
67
68

```



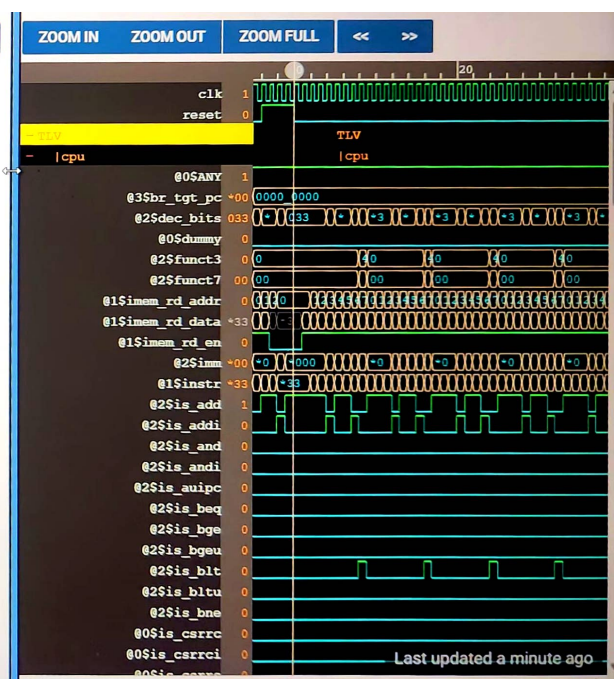
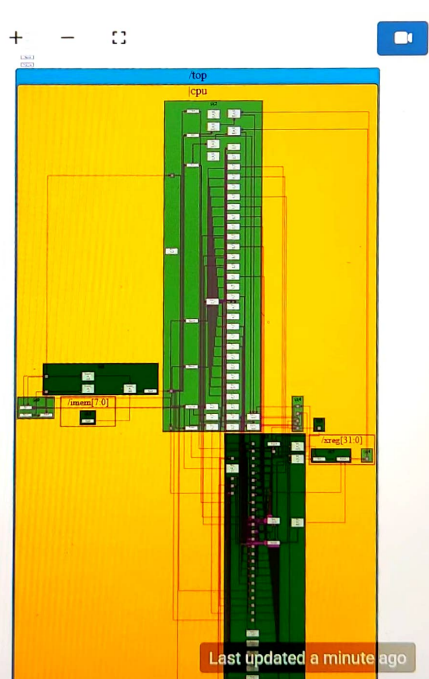
```
EDITOR NAV-TLV LOG  
m4_asm(ADD, r10, r0, r0)  
m4_asm(ADD, r14, r10, r0)  
m4_asm(ADDI, r12, r10, 1010)  
m4_asm(ADD, r13, r10, r0)  
  
m4_asm(ADD, r14, r13, r14)  
m4_asm(ADDI, r13, r13, 1)  
m4_asm(BLT, r13, r12, 111111111000)  
m4_asm(ADD, r10, r14, r0)  
  
m4_define_hier(['M4 IMEM'], M4_NUM_INSTR  
  
|cpu  
  @0  
    $reset = *reset;  
  
    $start = !$reset && >>1$reset;  
  
    $pc[31:0] =  
      >>1$reset ? 32'd0 :  
      >>3$staken_br ? >>3$br_tgt_pc  
      >>1$pc + 32'd  
  
  @1  
    $imem_rd_en = !$reset;
```



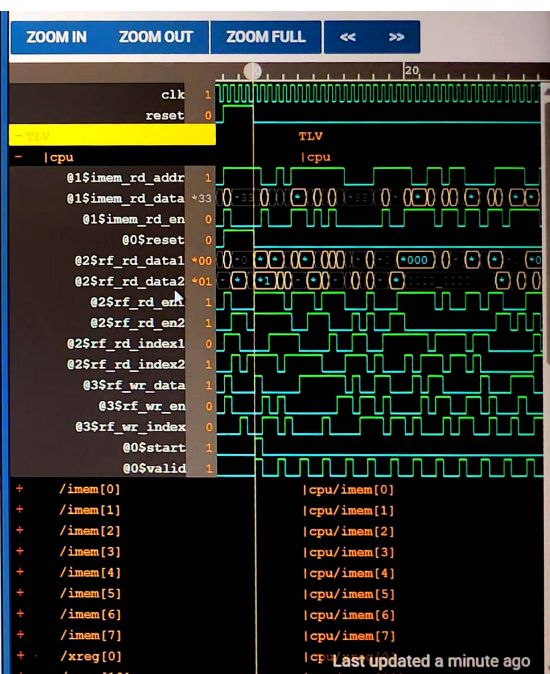
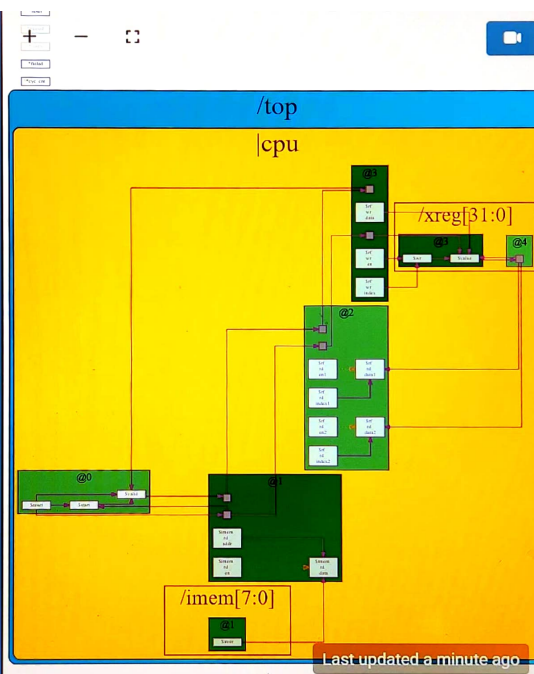
```

68 $src1_value[31:0] = (>2$rf_wr_en && (
69 $src2_value[31:0] = (>2$rf_wr_en && (>2$rd
70
71 $opcode[6:0] = $instr[6:0];
72 $funct3[2:0] = $instr[14:12];
73 $funct7[6:0] = $instr[31:25];
74
75 $dec_bits[10:0] = {$funct7[5], $funct3, $opc
76
77 $imm[31:0] =
78   {{21{$instr[31]}}, $instr[30:20]};
79
80 $is_beq = $dec_bits ==? 11'bx_000_1100011;
81 $is_bne = $dec_bits ==? 11'bx_001_1100011;
82 $is_blt = $dec_bits ==? 11'bx_100_1100011;
83 $is_bge = $dec_bits ==? 11'bx_101_1100011;
84 $is_bltu = $dec_bits ==? 11'bx_110_1100011;
85 $is_bgeu = $dec_bits ==? 11'bx_111_1100011;
86
87 $is_add = $dec_bits == 11'b0_000_0110011;
88 $is_sub = $dec_bits == 11'b1_000_0110011;
89
90 $is_and = $dec_bits == 11'b0_111_0110011;
91 $is_or = $dec_bits == 11'b0_110_0110011;
92 $is_xor = $dec_bits == 11'b0_100_0110011;

```



```
40 |cpu
41 @0
42 $reset = *reset;
43
44 $start = !$reset && >>1$reset;
45
46 $valid = !$reset && ($start || >>
47
48
49
50 // YOUR CODE HERE
51 // ...
52
53 // Note: Because of the magic we are
54 // be sure to avoid having una
55 // other than those specifical
56
57
58 // Assert these to end simulation (befo
59 *passed = *cyc_cnt > 40;
60 *failed = 1'b0;
61
62 // Macro instantiations for:
63 // o instruction memory
64 // o register file
65 // o data memory
```



```

38 |cpu
39 |@0
40 |$reset = *reset;
41 |
42 |$start = !$reset && >>1$reset;
43 |
44 |$valid = !$reset &&
45 |($start || >>3$valid);
46 |
47 |$pc[31:0] = >>1$reset ? 32'd0 :
48 |>>1$pc + 32'd4;
49 |
50 |@1
51 |$imem_rd_en = !$reset;
52 |
53 |$imem_rd_addr[M4_IMEM_INDEX_CNT-1:0] =
54 |$pc[M4_IMEM_INDEX_CNT+1:2];
55 |
56 |$instr[31:0] = $imem_rd_data;
57 |
58 |// YOUR CODE HERE
59 |// ...
60 |
61 |*passed = *cyc_cnt > 40;
62 |*failed = 1'b0;
63 |
64 |

```

